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Project      : Sales Analysis using SQL
File Name    : 03_insert_Data.sql
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Description :
This script populates all dimension tables and the fact table
from the cleaned sales dataset.

Steps Performed:
1. Load Product Dimension
2. Load Date Dimension
3. Load Region Dimension
4. Load Customer Dimension
5. Load Sales Representative Dimension
6. Load Sales Channel Dimension
7. Load Payment Dimension
8. Load Fact Sales Table
9. Recalculate Sales Amount

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-- STEP 1: Load Product Dimension
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INSERT INTO dim_product (product_category)
SELECT DISTINCT product_category
FROM sales_data;

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=====
-- STEP 2: Load Date Dimension
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INSERT INTO dim_date
(
    full_date,
    year,
    month,
    quarter,
    day,
    month_name,
    day_name

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)
SELECT DISTINCT
    sale_date,
    YEAR(sale_date),
    MONTH(sale_date),
    QUARTER(sale_date),
    DAY(sale_date),
    MONTHNAME(sale_date),
    DAYNAME(sale_date)
FROM sales_data;

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-- STEP 3: Load Region Dimension
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INSERT INTO dim_region(region)
SELECT DISTINCT region
FROM sales_data;

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-- STEP 4: Load Customer Dimension
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INSERT INTO dim_customer(customer_type)
SELECT DISTINCT customer_type
FROM sales_data;

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-- STEP 5: Load Sales Representative Dimension
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INSERT INTO dim_sales_rep(sales_rep)
SELECT DISTINCT sales_rep
FROM sales_data;

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-- STEP 6: Load Sales Channel Dimension
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INSERT INTO dim_sales_channel(sales_channel)
SELECT DISTINCT sales_channel
FROM sales_data;

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-- STEP 7: Load Payment Dimension
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INSERT INTO dim_payment(payment_method)
SELECT DISTINCT payment_method
FROM sales_data;

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-- STEP 8: Load Fact Sales Table
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INSERT INTO fact_sales
(
    product_key,
    date_key,
    region_key,
    customer_key,
    sales_rep_key,
    sales_channel_key,
    payment_key,
    sales_amount,
    quantity_sold,
    unit_cost,
    unit_price,
    discount
)

SELECT
    p.product_key,
    d.date_key,
    r.region_key,
    c.customer_key,
    sr.sales_rep_key,
    sc.sales_channel_key,
    py.payment_key,
    s.sales_amount,
    s.quantity_sold,
    s.unit_cost,
    s.unit_price,
    s.discount

FROM sales_data s

JOIN dim_product p
ON p.product_category = s.product_category

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JOIN dim_date d
ON d.full_date = s.sale_date

JOIN dim_region r
ON r.region = s.region

JOIN dim_customer c
ON c.customer_type = s.customer_type

JOIN dim_sales_rep sr
ON sr.sales_rep = s.sales_rep

JOIN dim_sales_channel sc
ON sc.sales_channel = s.sales_channel

JOIN dim_payment py
ON py.payment_method = s.payment_method;

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-- STEP 9: Recalculate Sales Amount
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SET SQL_SAFE_UPDATES = 0;

UPDATE fact_sales
SET sales_amount =
(unit_price * quantity_sold) * (1 - discount);

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-- STEP 10: Verify Loaded Data
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SELECT *
FROM fact_sales;

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